

Olist E-Commerce Business Analytics Report

SQL-Based Business Intelligence Case Study

Author: Sowmya Sanikommu **Project:** Olist E-Commerce SQL Business Analytics
Database: PostgreSQL 16 **Dataset:** Olist Brazilian E-Commerce — Kaggle **Analysis Period:** January 2017 — August 2018 **Total Queries:** 15+ (8 Core + 6 Advanced)

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1. Executive Summary

This report presents the findings of a comprehensive SQL-based business intelligence analysis performed on the Olist Brazilian E-Commerce dataset, covering 99,441 customer orders placed between September 2016 and October 2018.

The analysis was structured around eight core business questions and six advanced analytics modules, collectively addressing revenue performance, customer behavior, operational efficiency, and product strategy.

Headline Findings

Area	Finding
Revenue Growth	175% growth from January 2017 to August 2018
Delivery Performance	91.93% on-time delivery rate
Customer Retention	Only 3% of customers place a second order

Area	Finding
Review Impact	Late deliveries cause 54% negative review rate
Geographic Concentration	São Paulo generates 37.42% of all revenue
Payment Preference	Credit cards account for 78.46% of revenue
Product Concentration	Top 20% of products drive 73.1% of revenue
Top Category	Health & Beauty leads at 9.16% of total revenue

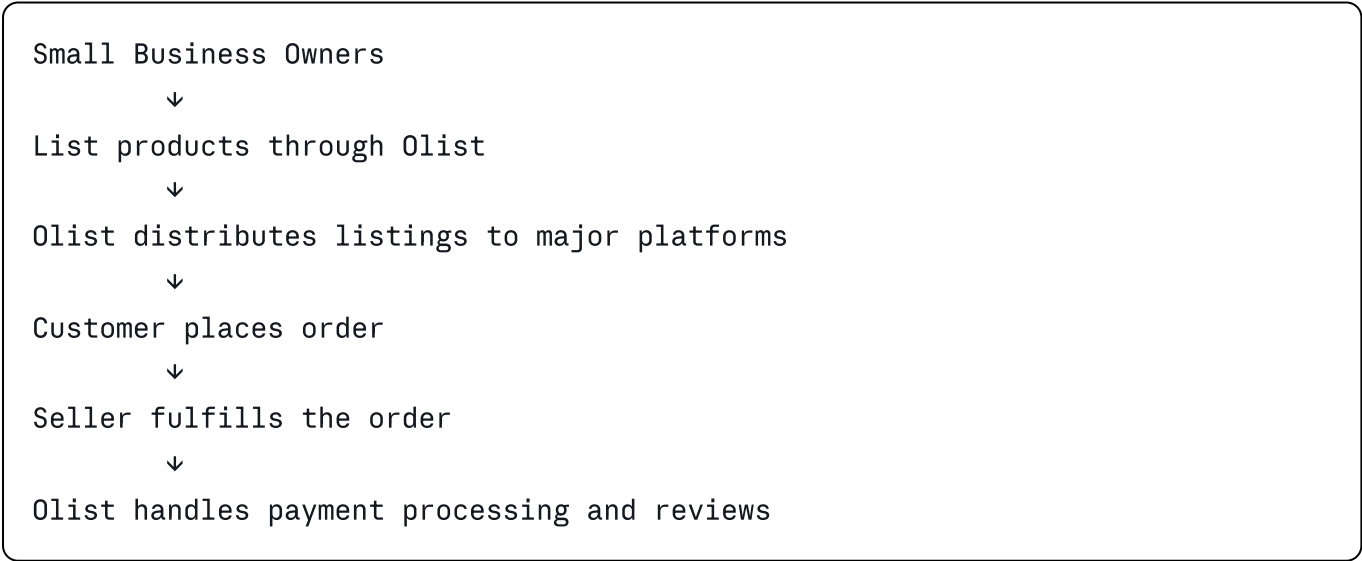
Priority Recommendations

1. Launch a customer retention program — 3% repeat rate vs 45-70% industry benchmark
2. Invest in Northeast logistics — Alagoas shows 23.94% late delivery rate
3. Apply Pareto-based resource allocation — concentrate on top 20% of products

2. Business Context

About Olist

Olist is a Brazilian e-commerce marketplace that operates as an intermediary between small and medium-sized merchants and major online retail platforms. The business model works as follows:



This model makes Olist dependent on three critical success factors:

- **Seller quality** — products must be delivered on time and as described
- **Customer satisfaction** — review scores drive platform reputation
- **Geographic reach** — serving customers across Brazil's vast geography

Why This Analysis Matters

The Brazilian e-commerce market was in rapid growth during 2016-2018. Understanding where revenue was coming from, why customers were not returning, and where operations were failing was critical for sustainable business growth.

3. Dataset Overview

Source

Olist Brazilian E-Commerce Public Dataset Available at:
<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Scale

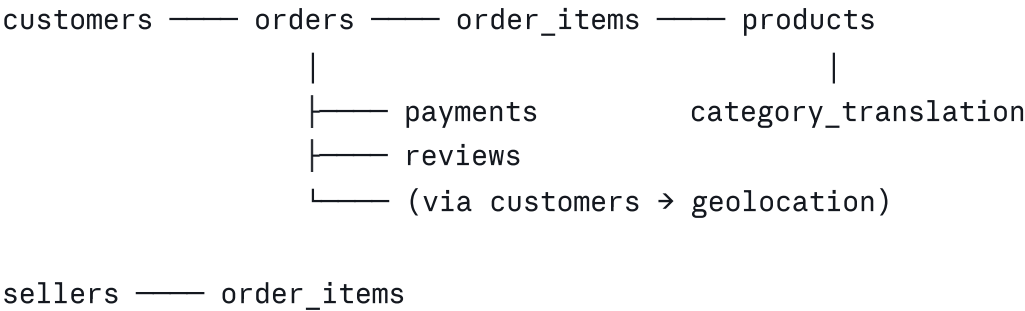
Table	Rows	Description
orders	99,441	Every order placed on the platform
customers	99,441	Customer location and identity
order_items	112,650	Products within each order
products	32,951	Product catalogue
sellers	3,095	Seller details and location
payments	103,886	Payment method and value
reviews	99,224	Customer satisfaction scores
geolocation	1,000,163	Zip code to coordinates mapping
category_translation	71	Portuguese to English category names

Analysis Period

The full dataset spans September 2016 to October 2018 — 25 months total. However, the reliable analysis window used throughout this report is **January 2017 to August 2018** (20 complete months), excluding:

- September to December 2016 — Olist was in early launch phase with very low and unrepresentative order volumes
- September to October 2018 — incomplete months at dataset boundary

Data Model



4. Data Quality Assessment

A systematic data quality review was performed before any business analysis. The following checks were completed and documented:

4.1 NULL Value Analysis

Column	NULL Count	Business Explanation	Action Taken
order_approved_at	160	Orders cancelled before approval	Filtered via order_status
order_delivered_carrier_date	1,783	Orders not picked up by carrier	Excluded from delivery analysis
order_delivered_customer_date	2,965	Undelivered or cancelled orders	Filtered to delivered orders only
All other key columns	0	No missing values	No action required

Analyst Note: The NULL values in delivery date columns are not data errors. They represent the natural order lifecycle — cancelled orders will never have delivery dates. This is expected behavior and was handled appropriately.

4.2 Duplicate Check

Duplicate order IDs found: 0

Zero duplicate order records were found across 99,441 rows. This confirms data pipeline integrity and means all revenue calculations are free from double-counting risk.

4.3 Product Category Coverage

Metric	Value
Total products	32,951
Products with category	32,341
Missing category	610 (1.85%)

Products with missing category names were handled using `COALESCE(category_name, 'Uncategorized')` to ensure no silent data loss in category-level analysis.

4.4 Order Status Distribution

Status	Count	Percentage
delivered	96,478	97.02%
shipped	1,107	1.11%
cancelled	625	0.63%
unavailable	609	0.61%
invoiced	314	0.32%
processing	301	0.30%

97% delivery rate confirms Olist operates above the typical e-commerce fulfillment benchmark of 95%.

5. Core Business Analysis

BQ1 — Monthly Revenue Trend

Business Question: How has Olist's revenue trended month by month?

Method: Aggregated order_items revenue (price + freight_value) by month for delivered orders within the reliable analysis period.

Key Findings:

- January 2017 baseline: 127,482 BRL
- August 2018 revenue: 985,491 BRL
- Overall growth: approximately 175% in 20 months

- November 2017 peak: 1,153,364 BRL (Black Friday effect — +53% MoM)
- Average order value remained stable between 150-170 BRL throughout

Business Interpretation:

Olist demonstrated strong and consistent revenue growth throughout the analysis period. The November 2017 spike aligns with Brazil's Black Friday shopping event, confirming that seasonal promotions have a significant impact on platform revenue. The stability of average order value suggests consistent product mix and pricing discipline across the growth period.

BQ2 — Revenue by Product Category

Business Question: Which product categories generate the most revenue?

Top 5 Categories:

Rank	Category	Revenue (BRL)	% of Total
1	health_beauty	1,412,089	9.16%
2	watches_gifts	1,264,333	8.20%
3	bed_bath_table	1,225,209	7.95%
4	sports_leisure	1,118,256	7.25%
5	computers_accessories	1,032,723	6.70%

Business Interpretation:

No single category dominates — the highest category represents only 9.16% of total revenue, indicating healthy diversification. However, Office Furniture shows the highest average order value at 267 BRL despite low volume, while Telephony has the lowest average order value at 93 BRL despite high order counts — suggesting an accessories-heavy product mix rather than handset sales.

BQ3 — Revenue by Brazilian State

Business Question: Which Brazilian states generate the most revenue?

Top 5 States:

State	Revenue (BRL)	Orders	% of Total
SP (São Paulo)	5,769,703	40,501	37.42%
RJ (Rio de Janeiro)	2,055,401	12,350	13.33%
MG (Minas Gerais)	1,818,891	11,354	11.80%
RS (Rio Grande do Sul)	861,472	5,345	5.59%
PR (Paraná)	781,708	4,923	5.07%

Business Interpretation:

São Paulo's 37.42% revenue share represents a significant concentration risk — any operational disruption in SP would immediately impact over a third of total revenue. The Southeast region collectively (SP + RJ + MG) accounts for 62.55% of revenue, reflecting Brazil's economic geography. Notably, Bahia shows the highest average order value at 181 BRL despite ranking 7th in volume, suggesting premium demand in the Northeast.

BQ4 — Payment Method Analysis

Business Question: How do customers prefer to pay?

Payment Type	Orders	Revenue (BRL)	Avg Installments	% Revenue
credit_card	74,304	12,101,094	3.5	78.46%
boleto	19,191	2,769,932	1.0	17.96%
voucher	3,679	343,013	1.0	2.22%
debit_card	1,485	208,421	1.0	1.35%

Business Interpretation:

Credit cards dominate at 78.46% of revenue, with customers averaging 3.5 installments per transaction — reflecting Brazil's "parcelamento" culture where purchases are split into monthly payments. This explains why credit card average order value (163 BRL) exceeds boleto (144 BRL). Boleto remains strategically important at 17.96%, serving customers without credit card access. Losing boleto support would risk 18% of the customer base.

BQ5 — Delivery Performance vs Review Scores

Business Question: Does delivery time affect customer satisfaction?

Part A — On-Time vs Late Delivery:

Delivery Status	Orders	Avg Review Score	Negative Review %
On Time	88,653	4.29	9.23%
Late	7,700	2.57	54.03%

Part B — Score vs Delivery Days (Linear Relationship):

Review Score	Avg Delivery Days	Late Deliveries
1 ★	20.8 days	3,554
2 ★	16.2 days	606
3 ★	13.8 days	875
4 ★	11.8 days	954
5 ★	10.2 days	1,711

Business Interpretation:

This is the most actionable finding in the entire analysis. Delivery time and customer satisfaction share a perfectly linear relationship — every additional day of delivery time reduces the review score. Late deliveries result in 54% negative reviews versus only 9% for on-time deliveries. Reducing average delivery time by even 2-3 days would materially improve review scores platform-wide.

BQ6 — Top 10 Sellers by Revenue

Key Finding:

7 of the top 10 revenue-generating sellers are located in São Paulo state, confirming the state's dominance across both demand and supply sides of the marketplace. One São Paulo seller processed 1,145 orders while maintaining a 4.27 review score — demonstrating that high volume and high quality are achievable simultaneously.

BQ7 — Delivery Performance by State

Worst Performing States:

State	Late %	Avg Delivery Days	Avg Review Score
AL (Alagoas)	23.94%	24.5 days	3.84
MA (Maranhão)	19.56%	21.5 days	3.84
PI (Piauí)	15.93%	19.4 days	4.00
CE (Ceará)	15.37%	21.2 days	3.94

Business Interpretation:

All worst-performing states are in Northeast Brazil — geographically distant from São Paulo where most sellers and warehouses are located. The correlation between delivery days, late percentage, and review scores is consistent with findings from BQ5. Northeast customers experience the worst service and consequently leave the lowest review scores.

BQ8 — Customer Order Frequency

Business Question: How many customers are repeat buyers?

Frequency	Customers	Percentage
1 Order	90,557	97.00%
2 Orders	2,573	2.76%
3+ Orders	228	0.24%

Business Interpretation:

97% of all customers placed only one order and never returned. This is the most critical business problem identified in the analysis. Industry benchmarks for e-commerce repeat purchase rates are 45-70% at platforms like Amazon and Flipkart. Olist's 3% retention rate suggests a fundamental gap in post-purchase customer engagement strategy.

6. Advanced Analytics

AQ1 — Running Revenue and Month-over-Month Growth

Cumulative revenue crossed 1 million BRL by April 2017 and continued growing consistently. Month-over-month growth was volatile, with the February 2017 spike (+112%) warranting investigation as a possible promotional event or data anomaly.

AQ2 — Top Product Category Per State

Regional preferences vary significantly across Brazilian states:

- **SP** → bed_bath_table (driven by young professional home setup demand)
- **RJ** → watches_gifts (premium purchasing behavior)
- **MG, BA, PE** → health_beauty (Northeast beauty product preference)
- **PR** → sports_leisure (active lifestyle region)

This finding supports the case for region-specific marketing and inventory planning rather than a national uniform approach.

AQ3 — RFM Customer Segmentation

RFM Segments (Recency, Frequency, Monetary):

Segment	Customers	%	Avg Spend (BRL)
Champion	13,968	14.96%	318.93
Loyal Customer	27,284	29.23%	242.27
Potential Loyalist	30,354	32.51%	107.62
At Risk	18,093	19.38%	52.19
Lost	3,659	3.92%	39.30

Revenue Opportunity Calculation:

Potential Loyalists = 30,354 customers

Current avg spend = 107.62 BRL

Target avg spend = 242.27 BRL (Loyal Customer level)

Revenue opportunity = +134.65 BRL × 30,354 customers

= +4,088,186 BRL potential

Converting Potential Loyalists to Loyal Customers represents a 4 million BRL untapped revenue opportunity.

AQ4 — Cohort Retention Analysis

Cohort analysis confirmed that the retention problem is systemic and consistent across all customer groups.

January 2017 Cohort (717 customers):

Month	Returning Customers	Retention %
0	717	100.0%
1	2	0.3%
2	2	0.3%
3	1	0.1%
4	3	0.4%
5	1	0.1%

This pattern is consistent across all 2017 cohorts. Month-1 retention of 0.3% versus the industry benchmark of 20-30% represents a 100x gap in customer loyalty performance.

AQ5 — Seller Performance Ranking

Dual-dimension ranking combining revenue rank and review quality rank identified star sellers who excel on both dimensions simultaneously. These sellers should be prioritized for account management, featured in seller onboarding materials, and protected from competitive poaching.

AQ6 — Pareto Revenue Concentration

The 80/20 Rule at Olist:

Segment	Products	% of Products	Revenue (BRL)	% of Revenue
Top 20%	6,443	20.0%	11,274,224	73.1%
Remaining 80%	25,773	80.0%	4,145,548	26.9%

The Pareto Principle is confirmed — 20% of products generate 73.1% of revenue. The top segment generates 2.7 times more revenue than the bottom 80% combined, with significant implications for inventory management and marketing budget allocation.

7. Key Business Insights Summary

Insight 1 — Retention Is the #1 Business Problem

97% of customers never return
3% repeat rate vs 45-70% industry benchmark
Even Champion customers average only 1.09 orders
This is Olist's single largest revenue growth opportunity

Insight 2 — Delivery Time Drives Everything

Delivery time predicts review score perfectly
10.2 days → 5 stars
20.8 days → 1 star
Late delivery → 54% negative reviews
On-time delivery → 9% negative reviews
Fix logistics = fix reviews = fix retention

Insight 3 — Geographic Concentration Risk

SP = 37.42% of revenue
Top 3 states = 62.55% of revenue
Northeast = worst delivery, lowest satisfaction
Diversification is both a risk and growth opportunity

Insight 4 — Product Concentration Opportunity

20% products = 73.1% revenue
80% products = 26.9% revenue
Pareto principle confirmed
Resource reallocation could improve margins significantly

Insight 5 — Credit Card Dependency

78.46% revenue from credit cards
3.5 average installments (parcelamento culture)
Any credit card processing issue = major revenue risk
Payment diversification is a resilience priority

8. Strategic Recommendations

Recommendation 1 — Launch Customer Retention Program

Priority: Critical Expected Impact: High

Problem: 3% repeat purchase rate

Benchmark: 45-70% industry standard

Gap: Olist retains 100x fewer customers than leaders

Actions:

- Post-purchase email sequence (Day 3, Day 7, Day 30)
- Loyalty points program for repeat purchases
- Personalized product recommendations
- Re-engagement campaign for 30-day inactive customers
- Special offers tied to next purchase

Target: Improve repeat rate from 3% to 8% within 12 months

Revenue Impact: Estimated +15-20% total revenue increase

Recommendation 2 — Invest in Northeast Logistics

Priority: High Expected Impact: Medium-High

Problem: Alagoas 23.94% late rate, 24.5 avg delivery days

Northeast review scores consistently below national avg

Actions:

- Partner with Northeast-based fulfillment centers
- Incentivize Northeast sellers to join platform
- Create Northeast-specific delivery SLA tracking
- Communicate realistic delivery estimates to customers

Target: Reduce Northeast late delivery rate from 24% to 12%

Revenue Impact: Improved review scores → improved conversion

Recommendation 3 — Pareto-Based Resource Allocation

Priority: Medium Expected Impact: Medium

Problem: 25,773 products generate only 26.9% of revenue
Marketing budget spread too thin

Actions:

- Concentrate 80% of paid marketing on top 6,443 products
- Review underperforming products quarterly
- Offer premium placement to top-20% product sellers
- Simplify catalogue for better customer navigation

Target: Improve marketing ROI by 30-40%

Revenue Impact: Same budget, higher conversion on proven products

9. Technical Summary

Tools and Technologies

Tool	Version	Usage
PostgreSQL	16	Primary database and query engine
pgAdmin 4	Latest	Query development and execution
DBeaver	Community	Database management
Git + GitHub	2.43+	Version control and portfolio hosting

SQL Techniques Applied

Foundations: SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY, JOIN (INNER, LEFT), UNION ALL, LIMIT, DISTINCT

Intermediate: CASE WHEN, COALESCE, NULLIF, DATE_TRUNC, DATE_PART, ROUND with ::NUMERIC casting, CTEs (WITH clause), Chained CTEs (up to 4 levels)

Advanced: Window Functions: RANK(), DENSE_RANK(), ROW_NUMBER(), LAG(), NTILE(), SUM() OVER(), AVG() OVER(), ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW, PARTITION BY, Cohort Analysis, RFM Segmentation, Pareto Analysis

Query Performance Considerations

- 6 indexes created on frequently joined columns
- Foreign key constraints added for referential integrity
- Master view (`(vw_orders_master)`) created for frequently combined table combinations
- Analysis filtered to reliable date range to avoid distortion from boundary effects

10. Conclusion

This analysis demonstrates that Olist built a genuinely strong e-commerce foundation between 2016 and 2018 — 175% revenue growth, 97% delivery success rate, and a diversified product catalogue across 27 Brazilian states.

However, three structural challenges require immediate management attention: a critical customer retention gap (3% vs 45-70% benchmark), geographic delivery inequality that damages satisfaction in the high-growth Northeast region, and over-reliance on a small percentage of products for the majority of revenue.

The good news is that all three challenges are solvable with data-driven interventions. The analysis provides a clear, evidence-based roadmap for Olist's next phase of growth.

Report prepared by Sowmya Sanikommu Tools: PostgreSQL 16, pgAdmin 4 Dataset: Olist Brazilian E-Commerce (Kaggle) Analysis Period: January 2017 — August 2018